INCH-POUND

MIL-P-71089 (AR)
AMENDMENT 4
14 September 1995
SUPERSEDING
AMENDMENT 3
09 May 1995

MILITARY SPECIFICATION

PROJECTILE, PRACTICE, 155MM, M804E1
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-71089(AR), dated 15 November 1991, and is approved for use by the U.S. Army Armaments, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Change "M804E1" to "M804A1"

1.1 Line 2: Change "M804E1" to M804A1"

2.1.1 Standard Military add "MIL-STD-1168 - Lot numbering of Ammunition"

PAGE 2

2.1.2 Drawings, 12913926: Change "M804E1" to "M804A1"

* Add drawing "9362569 - Packing & Marking for Pallet, Wood Ammo for PJCTL 155MM".

PAGE 4

3.8 Delete in its entirety.

Add "3.10.6.1 Rust on cup. At the time of receipt inspection and at assembly, the cup surface may exhibit surface rust, absent pitting and exfoliation. Loose rust shall be removed from the threads prior to asssembly.

PAGE 6

4.3.1 Line 6, Under "Quantity" delete "25" and substitute "5".

4.3.1 Line 7: Change "M804E1" to "M804A1"

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PAGE 7

Table I, Examination or Test: Change "M804E1" to "M804A1"

Table I, Delete the followg:

"(Except Proving Ground Test)" and
"Proving Ground Test 20, Level I, 3.8, 4.5.7"

PAGE 12

4.4.2.1 Major 104 Delete "20, 0-1" and substitute "4.4.3.2.2".

PAGE 22 and 23

4.4.2.10 Title block: Change "M804E1" to "M804A1"

PAGE 24

* 4.4.2.11 Delete drawing No "9331807" and substitute "9362569". Major 102, Delete "National Stock Number or DOD COde Missing or Unidentifiable and substitute "support cover missing".

PAGE 25

4.4.3.2.1.1 Add the following new sentence: "If the hardness of any cup is over the maximum requirement, that cup shall be rejected and removed from the lot."

4.4.3.2.1.2 Add the followin new sentence: "If the hardness of any sample cup is over the maximum requirement, all parts in that stress relief lot shall be tested for hardness, and all cups over the maximum requirement shall be rejected and removed from the lot."

Add "4.4.3.2.2 Tensile force. FIve sample cups shall be selected from each heat treatment or stress relief lot and subjected to the tensile force test. The sample cups shall be those with the lowest hardness, considered to represent acceptable processs, as measured in 4.4.3.2.1.1 or 4.4.2.1.2 above. If any sample cup fails to meet the applicable requirement, the heat treatment or stress relief lot represented by this sample, shall be rejected. Procedures for

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PAGE 28

4.5.7 Delete in its entirety

* 5.1 Delete in its entirety and substitute new paragraphs 5.1 and 5.2:
"5.1 Packing. Level A. The projectile shall be packed in accordance with dwg 9362569.
5.2 Marking. Marking shall be in accordance with dwg 9362569."

PAGE 29

6.3, Lines 3 and 4, Delete"Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: SMCAR-QAAR-I (D), Picatinny Arsenal, NJ 07806-50000." and substitute "Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000."

6.6, Line 4, Delete the following: "ballistic test data and"
Line 6, Delete: "SMCAR-CCH-P and SMCAR-QAR-Q (D)"
and substitute: "AMSTA-AR-FSA-M and AMSTA-AR-QAA-R."

6.8, Line 3, Delete "Dpvelopment and substitute "Development"

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Preparing activity:
Army - AR

(Project 1320-0016)

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assembly and Components	2 OF 2	See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Projectile Metal Parts Assembly			
	(Dwg. 12913927)			
	Examination for defects 1/	45, 100%	3.2	4.4.2.3
	Protective coating thickness 4/	45, 100%	3.10.2	4.5.4.1
	Salt spray testing 4/	5, 100%	3.2	4.5.4.2
	Functional test (Proving Ground)			
	(Dwg. 12913927)			
	Ballistic testing	10, 100%	3.11	4.5.8

NOTES:
1. The examination for defects are only for dimensional and visual inspections in paragraph 4.4.2.
4. After the dimensional and visual inspections of the 45 samples is completed, subject these same 45 samples to protective coating thickness, followed by salt spray testing of 5 of these samples.

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4.4.2.1 Sheet 2 of 2:

Drawing Number Block: Delete "12913429/12913928" and substitute "12913929"
Minor 216: Delete "Length from rear face to base diameter" and substitute "Length from rear face to basic diameter".

PAGE 17

4.4.2.2 Sheet 1 of 2:

Minor 202: Delete "Marking missing" and substitute "Marking (stamping) missing".

PAGE 18

4.4.2.2 Sheet 2 of 2:

Minor 214, Delete in its entirety.

PAGE 19

* 4.4.2.3 Title block, Delete "M804" and substitute "M804A1".

* 4.4.2.3 Add major 104:

"104", salt spray test, "4.4.3.5.1", "3.2", "4.5.4.2"

PAGE 20

4.4.2.4 Drawing Number block:

Delete "9331807/9334433" and substitute "12953254".

PAGE 23

4.5.2.1 a., line 2: delete "grinding to a sufficient depth" and substitute "grinding a flat to a sufficient depth"

PAGE 24

4.5.3.1.1 Line 2, 3 and 4: delete "prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 2.1) for test of 155MM, Proj., M107." and substitute "described in Drawing No. 9280429."

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4.5.3.2 Delete in its entirety and substitute the following:

"4.5.3.2 Mechanical properties and sidewall contact. The samples shall be sectioned by cutting through two planes normal to the longitudinal axis one centimeter forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for inspection for mechanical properties and the other section for inspection of sidewall contact."

PAGE 25

Figure 2, Delete the Title "Dynamic tear test bars." and substitute the following:

"Dynamic tear test bars. Approximate Location of Tension and Dynamic tear test bar specimens (T-Tensile, D-Dynamic tear)"

Add a note toward the right of figure and just above the horizontal centerline.

"Centerline of rotating band seat"

PAGE 26

* 4.5.3.2.1 Title: Delete title "mechanical properties" and substitute:

"Mechanical properties of disassembled rotating band"

* Add new paragraph 4.5.3.2.1.1:

4.5.3.2.1.1 Mechanical properties of annealed rotating band blanks. The band blank shall be cut from front to rear and flattened, without hammering which may further cold work the material, and the largest obtainable flat type specimen shall be cut from the band, the length of which shall provide a one inch gage length, plus adequate material at each end to accommodate the grips of the tensile testing machine. This method should yield a specimen with a 0.500 inch gage width. In any event, the gage cross section shall be determined to the nearest 0.001 square inch and utilized to calculate the tensile and yield strengths, and elongation.

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4.5.6 Hydrostatic test, line 4: Delete "3.6.3a" and substitute "3.7.3a".

4.5.6 Microstructure examination. Delete paragraph No. "4.5.6 and title and substitute the following: "4.5.7 Microstructure examination."

PAGE 29

* 6.7 Line 5, Delete "SMCAR-QAR-Q and SMCAR-FSA-SS" and substitute "SMCAR-QAR-R, SMCAR-FSA-MS and SMCAR-CCH-P"

6.7 Line 7: delete "61299."and substitute "61299-6000."

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